



TECHNICAL BINDER

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1360 PRESENTS



*Rendered On Fusion

HELIOS

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BINDER DESCRIPTIONS

Robot Name



Inspired by the Greek God of the Sun, ***“Helios”*** is a hero of Corgiopolis, as well as a gadgeteer who can be the star and orbit around its teammates simultaneously.



Binder Colours

Representing our team's colours, we used:

#0E1831 - Background Colour

#EA9637 - Used for titles/subtitles and links*

#FFFFFF - Used for regular text

* **Underlined Orange Text** is used in the document for hyperlinks, linking to the full Onshape CAD model for Helios.

GAME STRATEGY

Our first step in the design process for this game was an in-depth analysis of the three given archetypes of the competition, the strict limitations on each class would be the most influential factor on our design process. Within this analysis, we implemented a three-step system to rate each class.

1. Points Analysis



What are the maximum possible distinct point values each class can obtain given certain assumptions and ideal cycle times (approx. 11 seconds) based on our team's capabilities?



2. Team/Cycle Analysis

Which fits best in an entire team of different/the same archetypes?
How many cycles could the most optimal robot of this class score?

3. Mechanical Analysis



What subsystems would be used to optimize the given class of the robot? How did these subsystems fare in similar past games?

GAME STRATEGY

Points Analysis



The three classes all had different point values that could be scored in ideal autonomous and teleoperated scenarios. We tabulated the expected points scored from each class to determine which was best suited for us to play. We assumed that each class could score their preload capacity twice in autonomous (one cycle for gadgeteer), and assumed 10 cycles of each class's max preload capacity, given our team's inferred capabilities.

	Autonomous	Teleoperated
Commander	35 Pts: 2 Panels scored to fully obtain ownership (25 pts) Full ownership bonus (10 pts)	375 pts: Odd # cycles: 3 Panels for 1 FULL & 1 PARTIAL ownership (35 x 5 pts) Even # cycles: 3 Panels to complete FULL and score another FULL ownership (40 x 5 pts)
Mystic	42 Pts: 6 Bubbles scored to partially obtain ownership in 3 districts (30 pts) 6 Bubbles scored in NEUTRAL Districts (12 pts)	282 Pts: First 34 Bubbles: 6 Bubbles for 3 PARTIAL ownerships and NEUTRAL district scoring (36 x 5 + 24) Rest 26 Bubbles: Bubbles scored in PARTIAL districts (26 x 3)
Gadgeteer	45 Pts: 1 Panel, 2 Bubbles to fully own 1 district IN AUTO (35 pts) 3 Bubbles scored; 2 NEUTRAL, 1 FULLY OWNED (10 pts)	300 Pts: 1 Panel, 2 Bubbles to fully own 1 district (25 x 10 pts) 3 Bubbles scored; 2 NEUTRAL, 1 FULLY OWNED (5 x 10 pts)

Although this method did not consider de-scoring, tower climbs or alliance decisions, it provided us with a good perspective on the various scoring abilities of each class, and how they can impact the team overall. Through the analysis, we leaned more towards the **Commander** and **Gadgeteer** classes, however we still needed further analysis to complete our selection for class.

GAME STRATEGY

Team/Cycle analysis



1. Scoring Limitations

Our overall strategic goal is to be captain of alliance 1-3 to ensure that we would be paired with competitive teams, as historically these are the alliances that win events. Therefore, we wanted to be able to secure full ownership of districts on our own, to ensure that we don't lose valuable ranking points if paired with weaker teams during the group stage of competition. This pointed us in the direction of choosing the **Gadgeteer** and **Commander** classes.

2. Other Team Predictions

Immediately we anticipated that the Mystic class would be a favorite amongst inexperienced teams, given the simplicity and large historical database (2022, 2024, etc.) of mechanisms used to manipulate small balls. Thus, we decided that the ability to efficiently score story panels would be critical to ensure that our alliance can gain and defend full ownership of multiple districts. For more experienced teams likely being Gadgeteer and Commander, we would likely still pick Gadgeteer for its all round success and ability to fill the gaps instantaneously.

3. Versatility and Adaptability

As the game is quite competitive for district ownership, we wanted a bot that could also effectively de-score or switch district ownerships while cycling fast. Combining the quick scoring from speech bubbles and fast de-scoring through the panels, we felt the **Gadgeteer** class would be best equipped to adapt to different alliance classes, as well as effectively playing both offence and defense.



GAME STRATEGY

Mechanical Constraints



1. Frame Perimeter and Max Height

We immediately defined frame perimeter and max height constraints as a vital factor in choosing which class we would select. Choosing the Mystic class was a clear **disadvantage** as its significantly smaller frame (104" vs 120") and max height (42" vs 60") meant that we would have to compromise on the complexity of subsystems like ground intakes, climbers and shooters.

2. Weight Constraints

Additionally, we found the weight constraints to be a large consideration for the class we would choose. Having multiple different heavy subsystems on the robot would naturally favour a class that could carry more weight, which was the Commander class (125 vs 100 lbs). Although the Commander class could carry more weight, the heavier weight and nature of the panels would also offset this advantage, which displayed the need to lightweight our robot.

3. Game Piece Configuration

Through our initial strategy, we felt the story panels would be a difficult game piece to store on the robot in bulk (>1), which **deterred** us from the Commander class. The knob on the end of each panel presented an unorthodox challenge for us, as they would either require greater precision or more space on the robot for all of them to fit, much different to a similar-looking game piece from the 2024 game "CRESCENDO".



GAME STRATEGY

Final Choice

Culminating all considerations from our previous analysis, we decided to pick the **Gadgeteer** class. Our decisions were based off the 3-part analysis, which conclusively resulted with the gadgeteer to be the best class.

Points Analysis



In our points analysis, the Gadgeteer showed the promising independent autonomous results, which was a large consideration as it would be difficult for us to coordinate autonomous paths with alliance members. Although in teleop scoring the Gadgeteer fared slightly worse, its scoring was still quite strong, and paired with the other aspects made it a powerful consideration.



Team/Cycle Analysis

The team and cycle analysis heavily favoured the Gadgeteer class, where it could independently gain full ownership through 1 panel and 2 balls. Since the Commander class could not score any confirmed points (panel points could be overturned via loss of ownership), the Gadgeteer could score confirmed points while gaining full ownership, making it an overall better pick. The adaptability of the Gadgeteer is also unmatched, where it scores both game pieces, allowing it to play effectively to the instantaneous needs of the team.

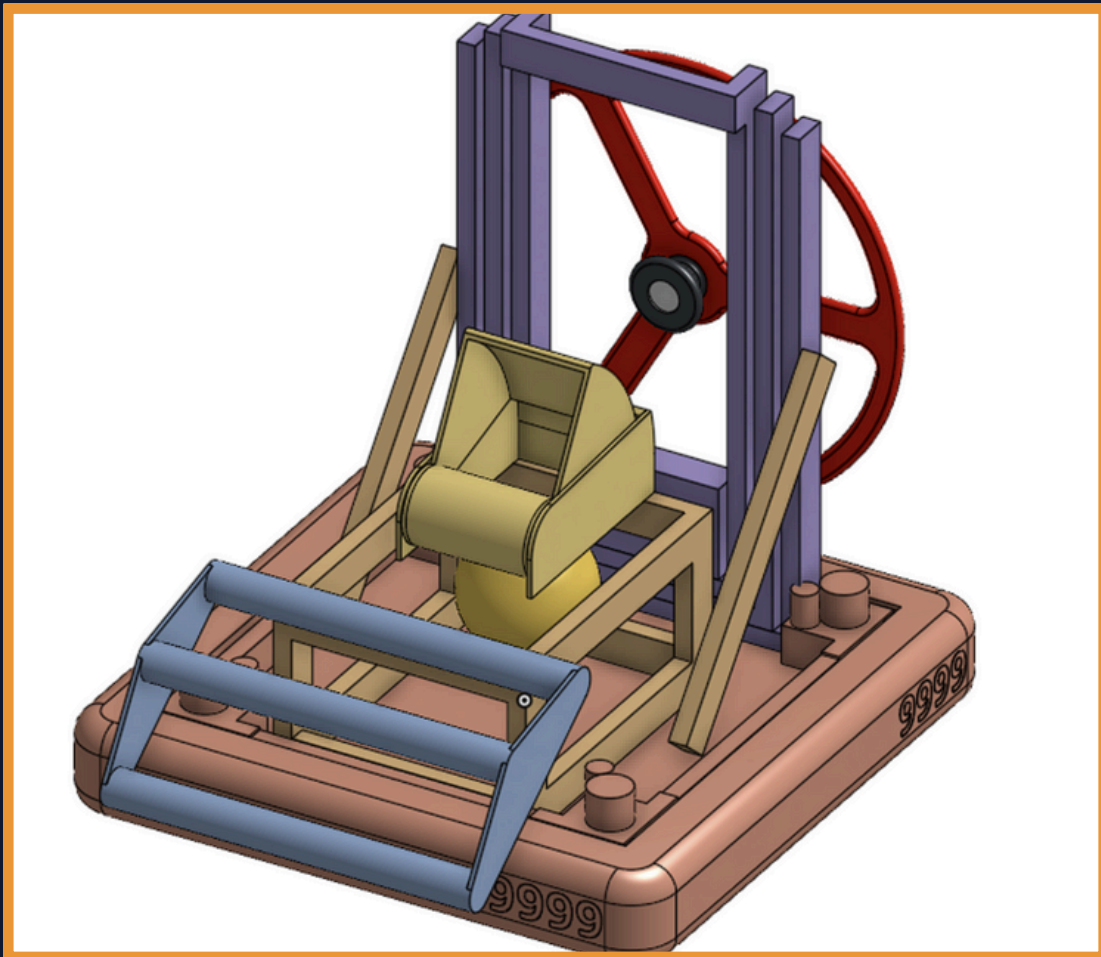
Mechanical Analysis



The Mechanical Analysis proved difficult for all classes, however ultimately favoured the Gadgeteer for its overall balanced mechanics. Although the Commander class had higher constraint limits, the difficulty in holding and placing multiple story panels ultimately ruled Commander out. Although the Mystic class required less mechanically, it's inability to extend to above 42 inches did not meet with climb requirements, which we anticipated Einstein level robots would need. The Gadgeteer class ultimately provided us with a safe height margin to effectively climb, as well as using similar mechanisms for us to hold the panels and bubbles simultaneously.

DESIGN PROCESS

KrayonCAD



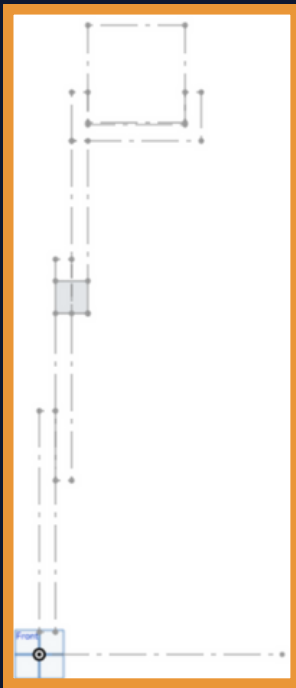
Initial KrayonCAD of our robot.

DESIGN PROCESS

Elevator



As the primary consideration in the selection of the Gadgeteer class was the ability for our robot to manipulate story panels, our first step in the design process was outlining the mechanism needed to do so. Since we wanted to eventually incorporate a climb given its points potential, we wanted to create an elevator that is tall enough to reach the climbing bar with a carriage that allows for the incorporation of a story panel end effector.

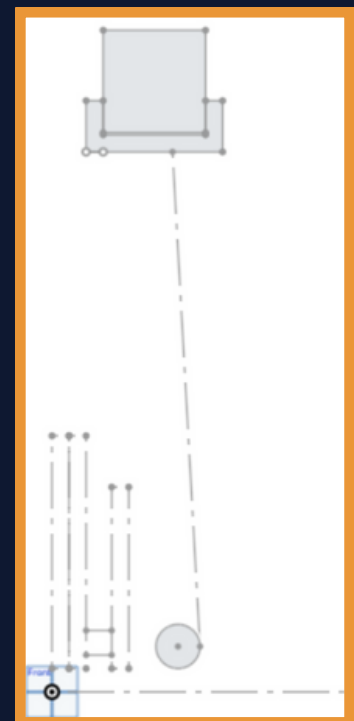


A first design that was discussed was an open top, continuous elevator with a tall carriage that allows the robot to reach the climb height. Eventually, we decided against this as:

- A three stage elevator was needed to reach the desired height; this would increase the complexity and weight of our mechanism.
- The height restrictions of our chosen class meant that the extension of our elevator would be severely restricted during normal play, limiting the height that the carriage could reach.

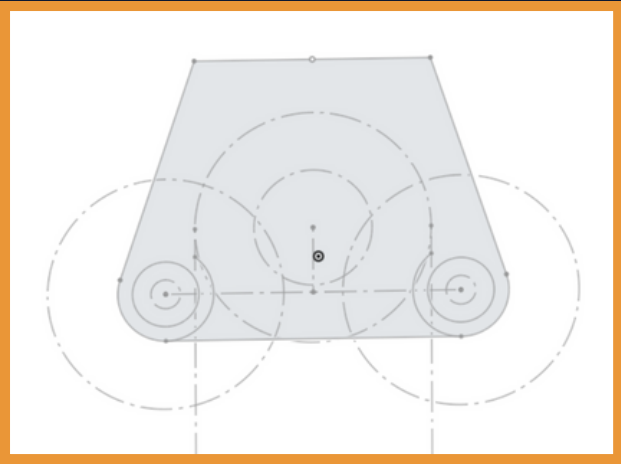
Our chosen design was a two-stage continuous elevator that utilized a detachable climbing mechanism inspired by team 118's 2016 climber and the physics behind rock-climbing cams. Its advantages were:

- Allows for **high climb** as it pulls the robot all the way up to the beam from the winch
- **Doesn't inhibit** normal elevator range when not climbing as the elevating mechanism isn't always attached to the elevator
- **Lighter** than other options of having an entirely separate elevator to get the climber up to there



DESIGN PROCESS

End Effector



Early on, we decided that cycle times would be arguably the most important part of this game as the ownership gaining and opponent de-scoring speed would be the biggest strategic tool in winning Einsteins. Thus, our story panel end effector must be:

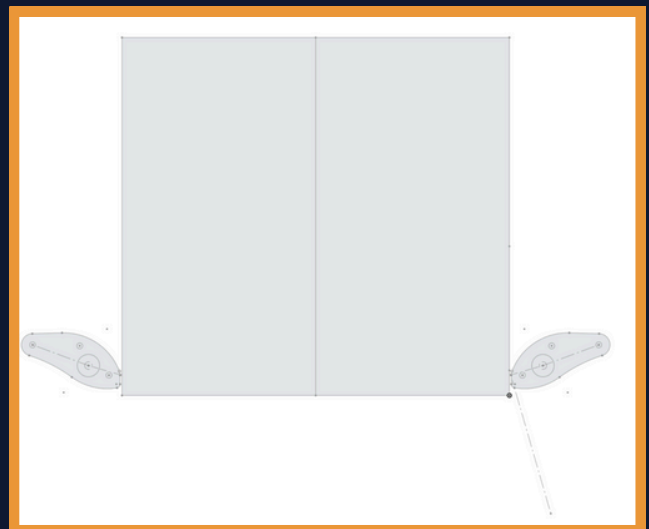
- Able to pick up from the **ground**
- Powered with **motors** to ensure fastest possible piece ownership
- Fixed on a **pivot point** on the elevator

Climber

Since the high climb had such a significant impact in comparison to in-game scoring, we knew that an effective climb could help us beat strong alliances during playoffs.

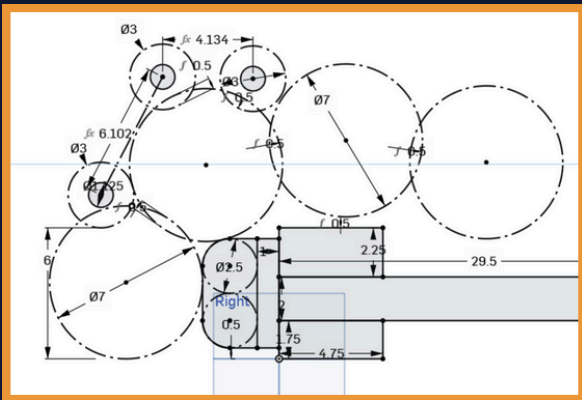
Thus, our climber must be:

- Detachable to ensure that our robot doesn't **exceed in-game height limit**
- Around **10-12 seconds** so other teams have less time to lower our ownership
- As mechanical as possible to **avoid electrical breakdowns** and save weight



DESIGN PROCESS

Intake



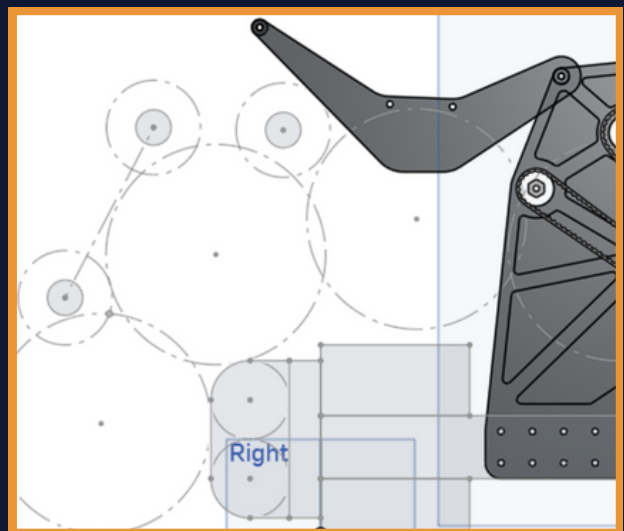
To maximize cycle times, we also wanted to intake speech bubbles from the ground. Our design considerations for this mechanism were:

- It must be strong enough to **take hits** from other robots
- It must be powered from **inside** the robot
- Must be **as wide as possible** to allow for optimized intaking
- Must be able to **deploy and retract quickly**
- Should be **light** to maximize weight efficiency

Indexer

Design considerations:

- Had to be compact due to limited space for ball scoring mechanism
- **Light** to maximize weight efficiency
- Be able to hold **three balls** comfortably, four possibly
- Quick indexing



SUBSYSTEMS

Elevator



- **2-Stage** continuous elevator with a starting height of 29.5 inches & max height of 52 inches
- Powered by **2 Kraken X44** on a 3.667:1 gear reduction
- Deploys to full height in **0.48 seconds**
- Supported by **2 1x1 boxtubes** and **1 tie rod** (other space taken up by climber)

Arm

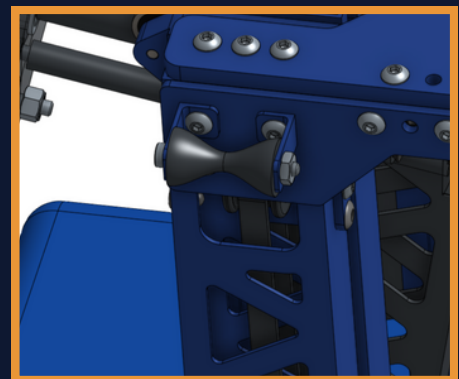
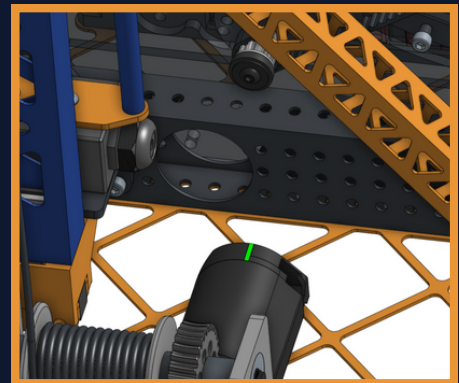
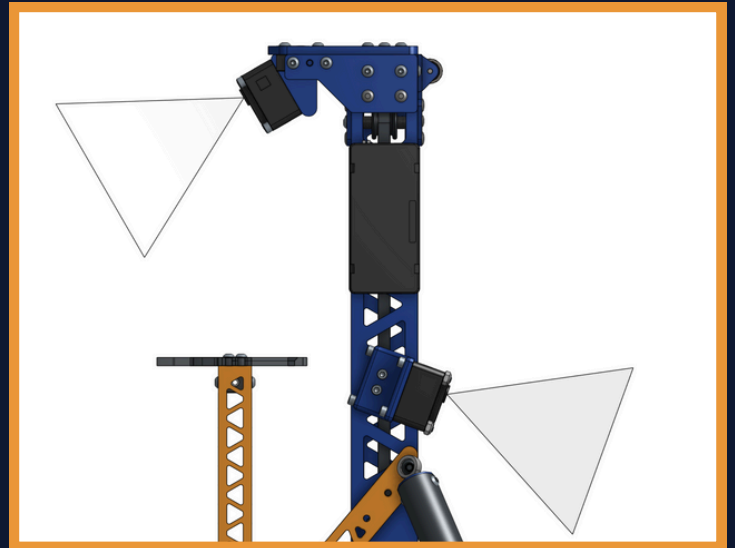
- **36:1** planetary gearbox with a **Kraken X60** mounted on the elevator carriage for arm pivot
- Wheels are powered by a **Kraken X44** at a **8:1** reduction



SUBSYSTEMS

Electrical

- **PDH** sits under shooter as central hub for power distribution, with **VRM, PDH, and Main Breaker** placed on open robot surfaces for easy pit checks
- **Two CANRange beam break sensors** positioned in arm and indexer, and **WCP through-bore encoder** mounted on arm pivot
- Radio is mounted high on elevator to meet preferred placement requirements and **strong connectivity**
- Battery acts as a **ballast** below the indexer during climbs, with all programming and power loads verified through **ReCalc** for load-bearing subsystems
- Two Limelight 4s for **AprilTag** vision and object detection for auto/teleop alignment
- Wire pathing and weight were taken into consideration (elevator cable + cable pathing in drive train)

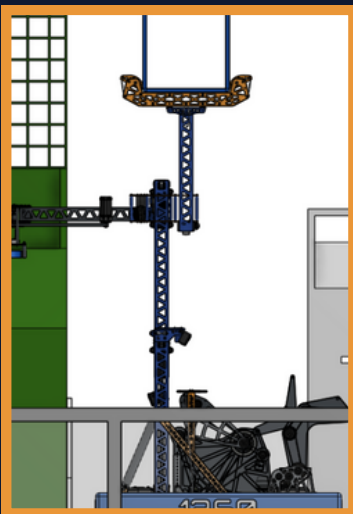


SUBSYSTEMS

Climber



The final climbing mechanism was inspired by team 118's 2016 detachable climber and conventional rock climbing cams. The actual climbing mechanism resembles a trident and remains disconnected from the robot throughout gameplay to adhere to height restrictions, then attaches to the elevator via an 160 pound electromagnet during endgame. With this, the flat portion of the climber can reach a max height of about 65.5 inches. Then, a winch pulls down and rotates the cams on each side of the trident, resulting in a very strong frictional hold. The winch continues operating, pulling the robot up after the cams clamp on.



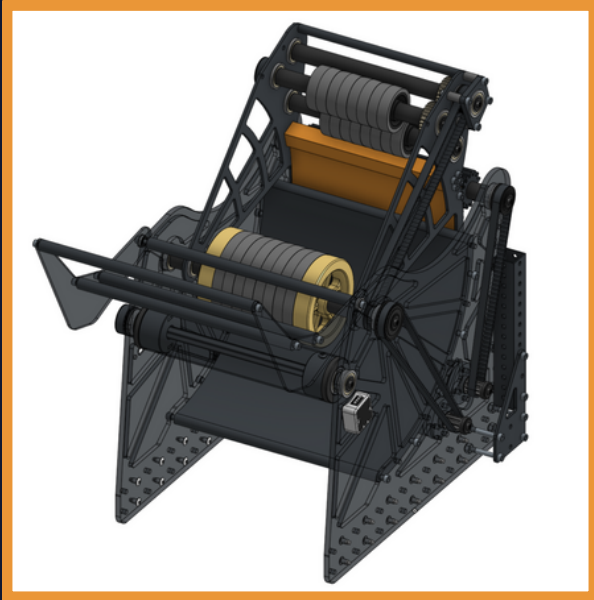
Maximum extension of the robot with electromagnet activated.



Deployment of the cams to grip the climbing bar.

SUBSYSTEMS

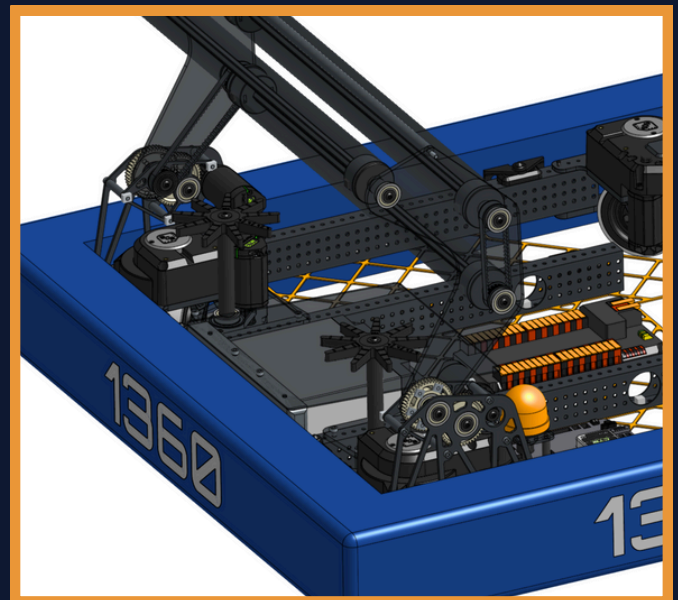
Shooter



- **4** inch flywheels
- kick-up roller to stop balls and give time for flywheel to accelerate
- **3 Kraken X44** to power kick up roller, flywheels and hood
- **60 degrees** of hood rotation

Indexer

- **2x4** inch diameter Star Wheels for centering balls: Powered by one **Kraken X44**
- Spring-loaded top hard stop to **prevent** balls from going up (inspired by 4414's 2020 offseason)
- Hard stop is partly pushed back by intake as it retracts to not block the shooter
- **Four** ball capacity



SUBSYSTEMS

Intake



- Slap down intake powered by a **Kraken X44**
- Wheels powered by a **Kraken X44** and belts
- Gearbox for both located on the robot - > no motors on the intake
- Made from **polycarbonate** to be strong while still being light



Drivetrain

- To maximize speed, **four Kraken MK5N Swerve Modules** were used
- Final frame dimensions of **29.5 inches by 29.5 inches**
- The bellypan of the robot was lightened to stay under weight limits.

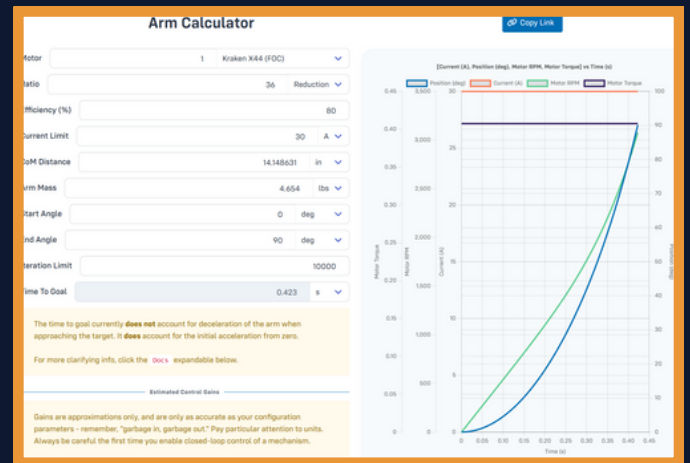
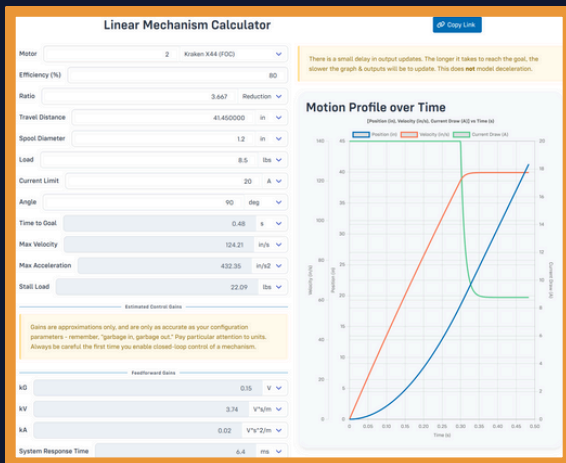


SUBSYSTEMS

Calculations

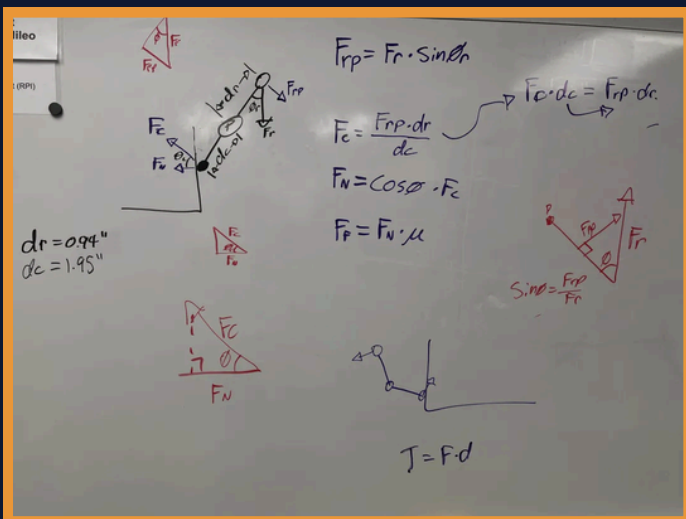
Elevator & Arm Gearbox Ratios

justified with proper weights, current limits, and weights.



Dynamics/Mechanics Calculations

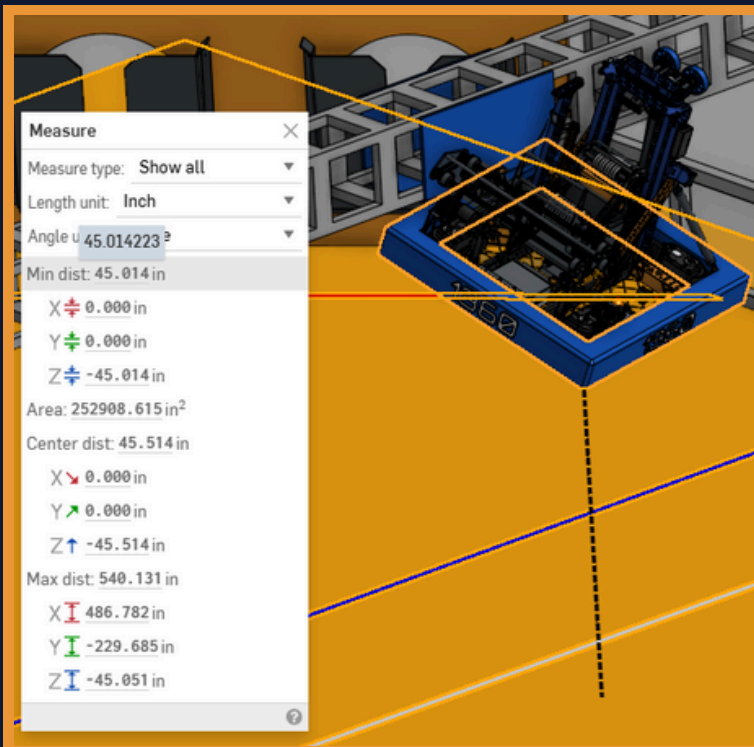
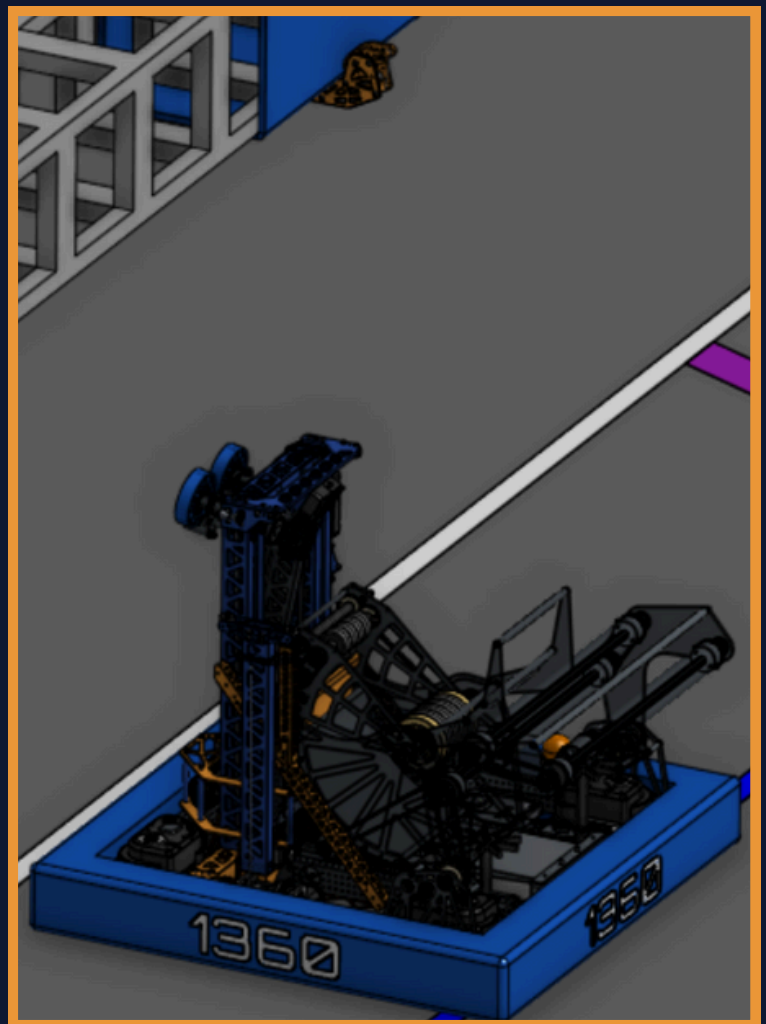
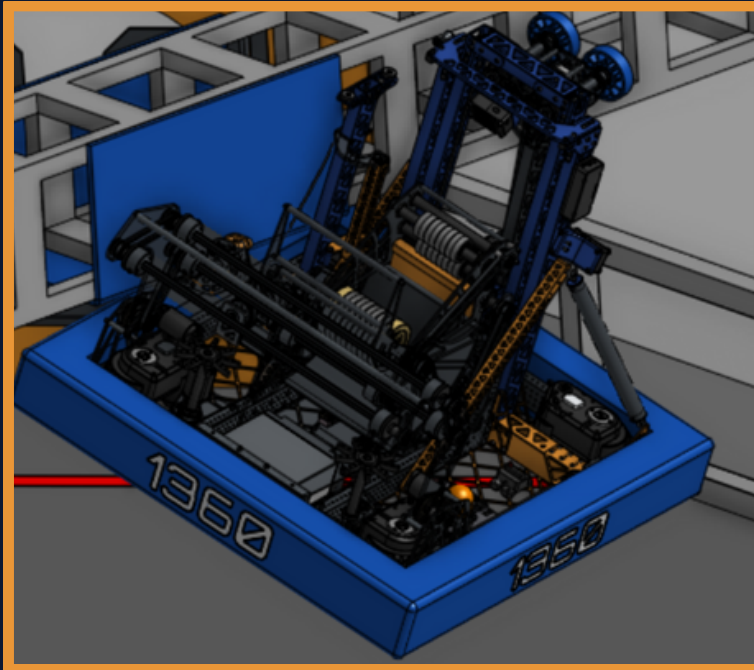
Justification for the strength of the friction force outweighing the strength of the gravitational force through free-body diagrams and related calculations.



Frp	Fc	Fn	Friction Force	
418.45	676.21	347.46		469.08
Angle Rope	Angle Normal	Rope to Pivot	Pivot to Contact	Coeff Friction
70.15	59.08	2.00	1.24	1.35
1.22	1.03			
Robot Force	444.89			

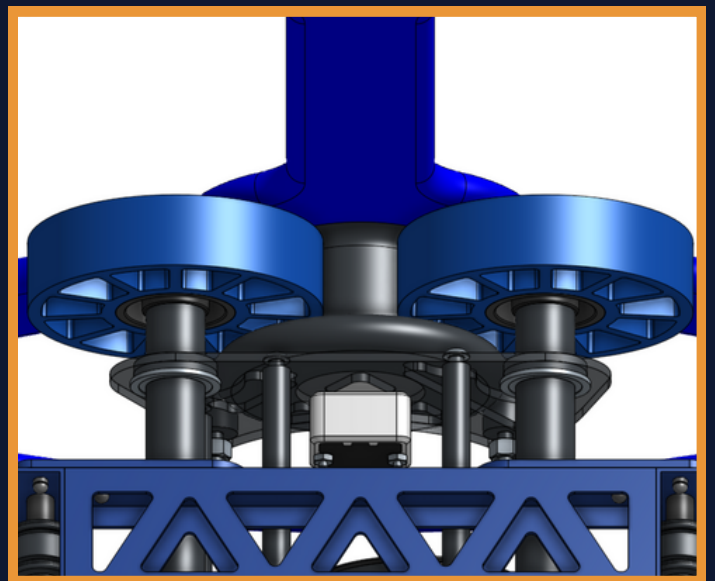
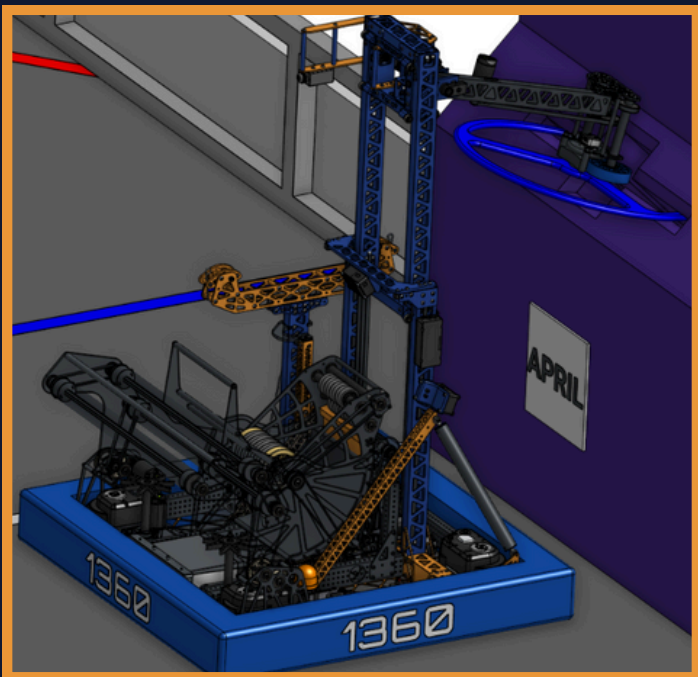
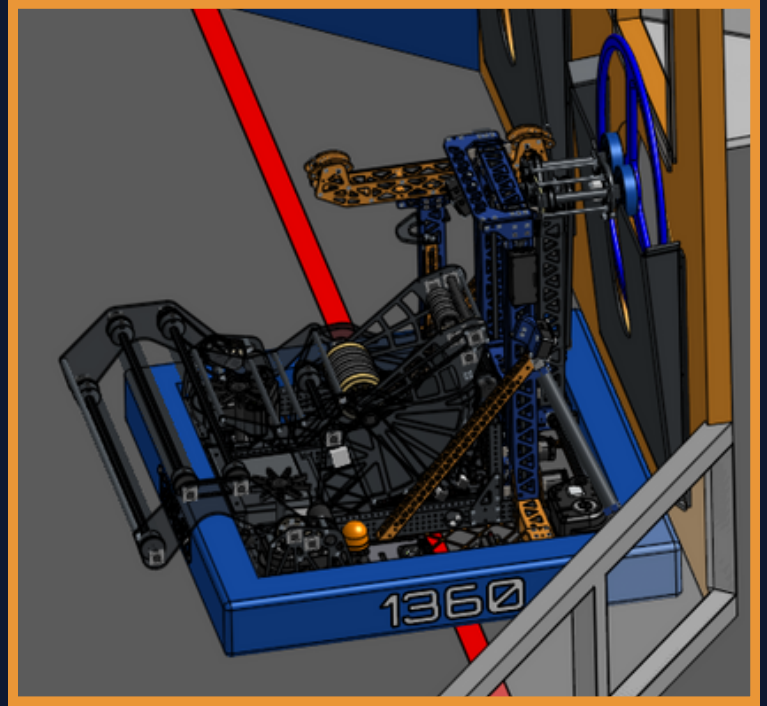
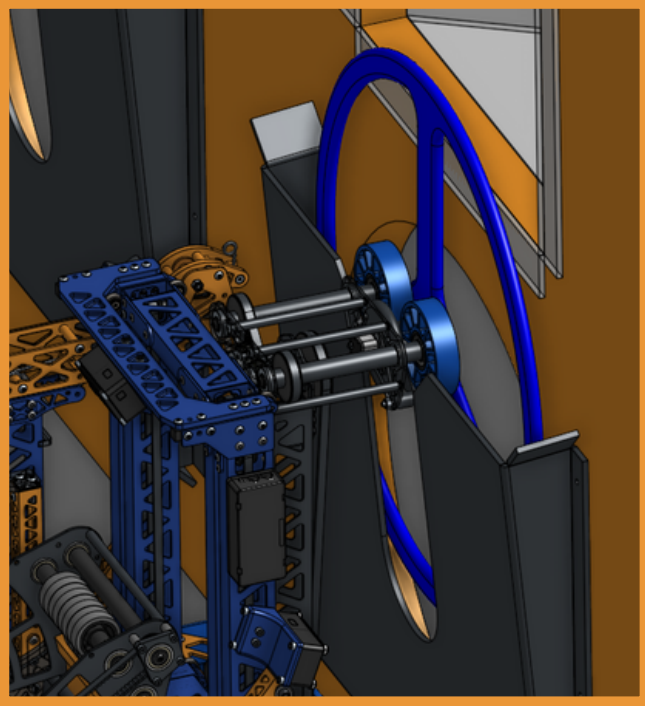
POSITIONS

Climb



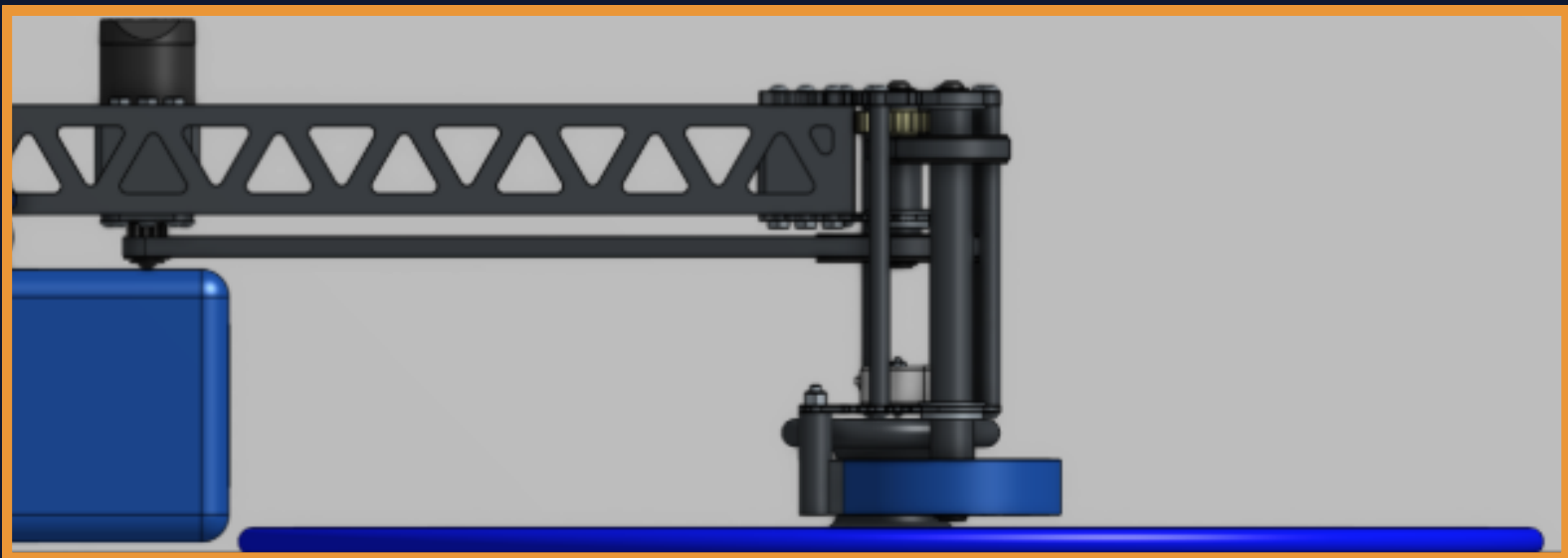
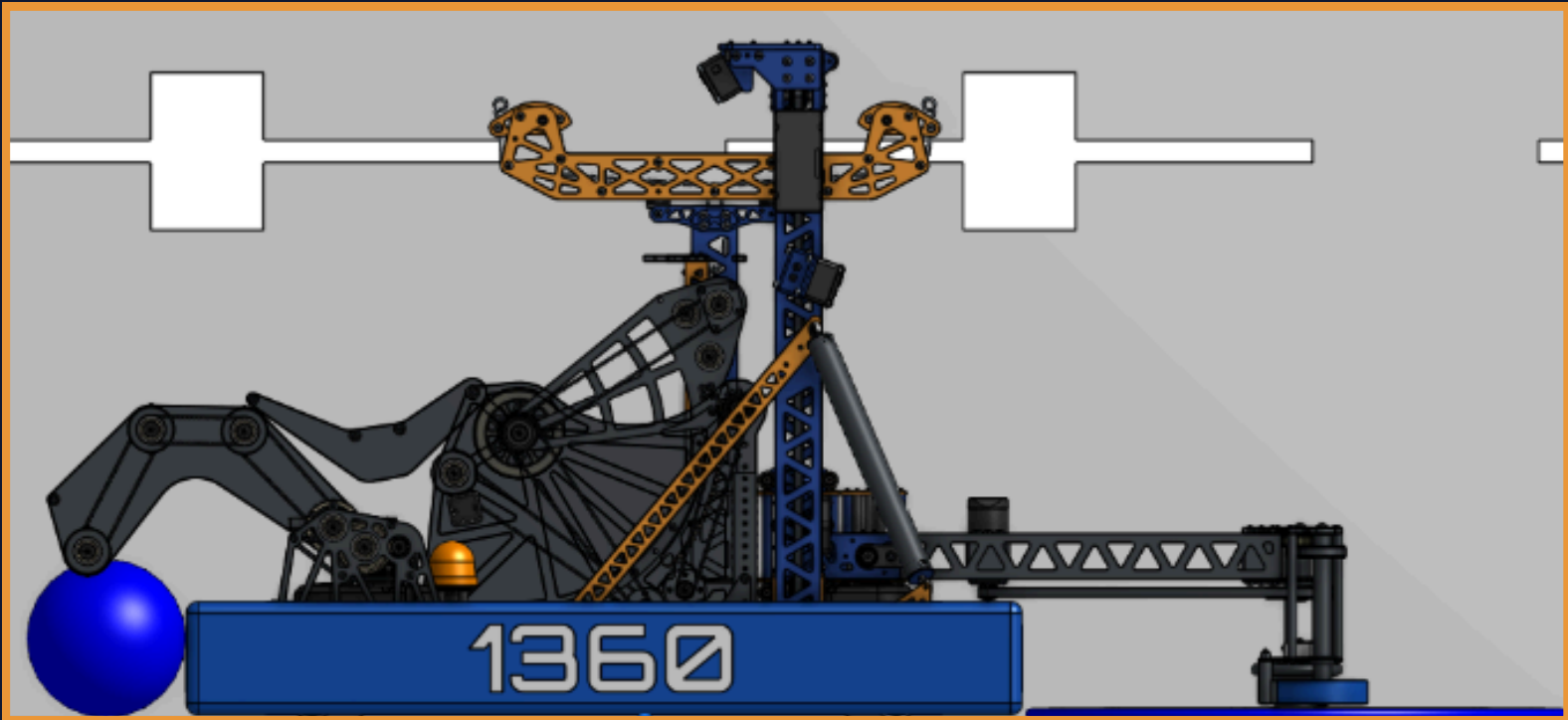
POSITIONS

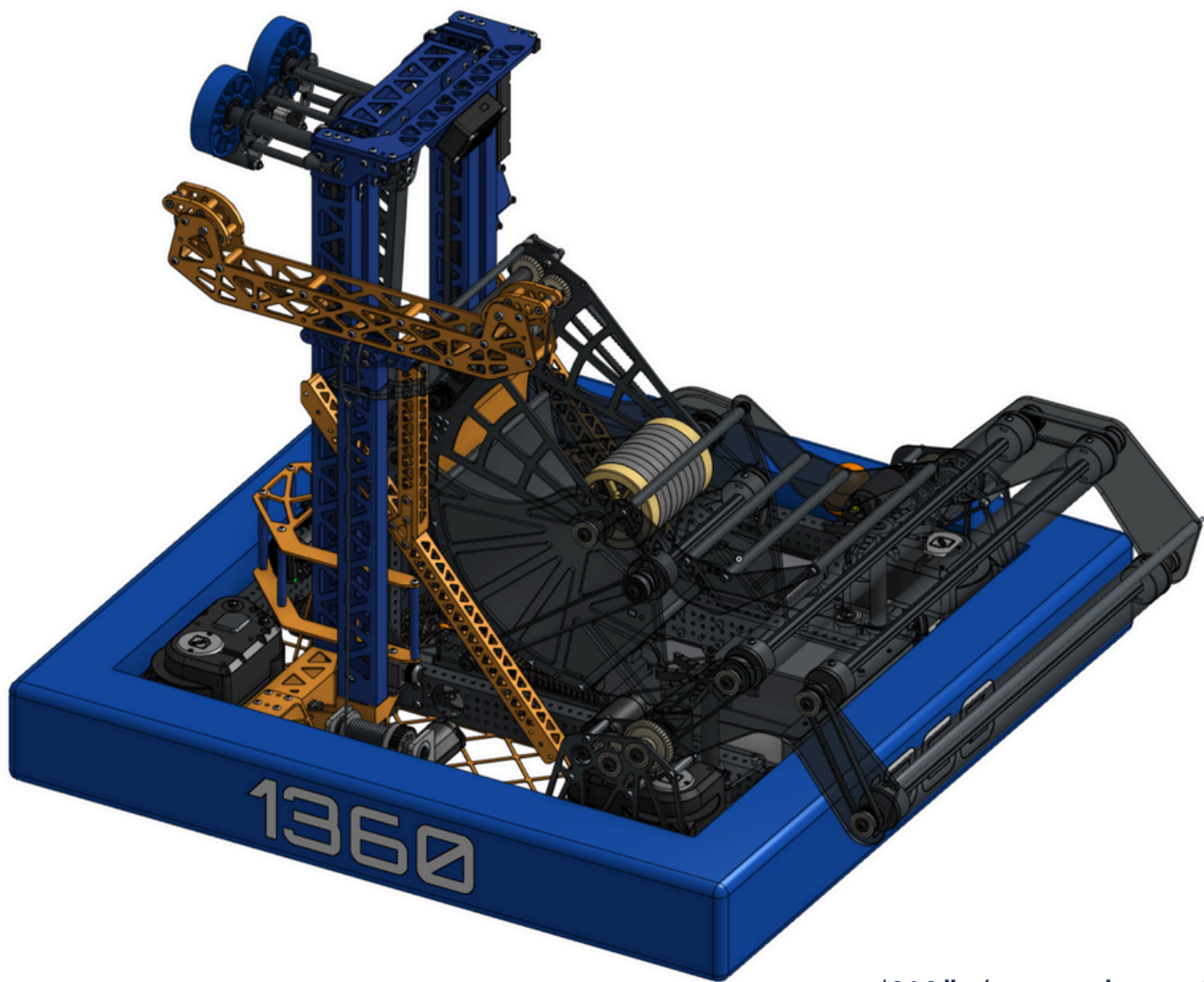
Disc Scoring



POSITIONS

Intake





*94.2 lbs (as per onshape estimates)

FULL ROBOT CAD